

1) Identify the Problem Statement

Objective: Prove the problem exists, how painful it is, and who feels the pain.

Deliverables (what each team must submit)

- One crisp problem statement (1–2 sentences).
- A 300–500 word context summary with evidence (stats, quotes, citations or screenshots from interviews).
- At least **5 primary research items** (user interviews, survey responses, observational notes) and **3 secondary sources** (industry reports, articles, market data).
- Problem severity scorecard (quantify time/money/worse outcomes).

How to write a strong problem statement (template)

“[Target user] currently struggles with [specific pain], which causes [quantified harm: time lost, money lost, risk] because [root cause].”

Example:

“Smallholder farmers in rural Kenya currently lack affordable, real-time crop disease detection; this leads to up to X% crop loss per season and lost revenue because access to diagnostics and agronomy advice is limited.”

Research & evidence (practical steps)

- **Surveys:** 10–50 respondents in target group; include 3–5 quantitative questions (frequency, cost, time lost) + 2 open comments.
- **Interviews:** 5–10 recorded interviews (15–30 mins) with representative users; extract 3–5 direct quotes.
- **Observational:** field notes, screenshots, or videos illustrating the user workflow.
- **Secondary:** cite at least 3 reputable sources (industry blog, academic paper, NGO report, government stats).

Validation checklist

- Problem is specific (not “people want better apps” but a distinct pain).
- Evidence shows it affects a measurable group and has real cost/consequence.
- At least one documented user quote or case showing the pain.

- Demonstrates urgency / timing (why solve now).

Grading criteria (suggested)

- Clarity (20%), Evidence quality (35%), Research breadth (25%), Relevance (20%).

2) Define the Solution / Product

Objective: Describe exactly what you will build, how it works, and why it's the best realistic choice.

Deliverables

- One-line product pitch (tagline).
- 250–400 word product description explaining the core functionality and user flow.
- MVP feature list (must vs nice-to-have).
- Unique Value Proposition (UVP) — 3 bullet points why it's different/ better.
- Quick wireframes or screenshots (2–4 images) showing core screens or flows.

How to structure the solution explanation

1. **What it is** — product type (mobile app / web platform / API / desktop tool / hybrid).
2. **How it works (high level)** — 3–6 steps of user flow.
3. **Core features (MVP)** — features the team will deliver for their demo.
4. **Why it's viable now** — technology readiness, market timing, dependencies.

MVP feature list (example layout)

- **Must-have:** user onboarding, core action (e.g., upload photo → AI result), results page, basic analytics, user feedback.
- **Nice-to-have:** social sharing, payment gateway, advanced analytics, multi-language.

Differentiation / defensibility

- Cite one competitor and describe *how you're different* (faster, cheaper, localized, offline mode, better UX, partnerships).

- If using technology like ML or blockchain, explain why that tech is necessary and what barrier it creates for others.

Validation & prototyping

- Prototype test with at least 5 users (record task completion rate and qualitative feedback).
- If AI/model-based, show a 1–2 page summary of training/test dataset and accuracy metric (even if small).

Acceptance criteria

- MVP implements at least one core use case end-to-end.
- UVP is clear and defensible.
- Prototype test shows >60% success for core task or clear plan to reach it.

1) DEFINING USER PERSONAS

A **persona** is a research-based fictional representation of a user segment.
Each persona should be built from actual data (surveys, interviews, market research).

Type	Purpose
Primary Persona	The <i>main</i> user whose problem you are solving.
Secondary Persona	Users indirectly involved or benefiting, but not the main focus.
Edge Persona	Unusual or extreme users who test the limits of the product's usability or adaptability.

Persona Template (for students to complete)

Field	Description / Example
Persona Name	Give them a relatable first name, e.g., <i>Amina the Farmer</i> or <i>Kevin the Retailer</i> .
Photo/Icon	Use stock image or icon to humanize the persona.
Demographics	Age, gender, location, education, occupation, income range.

Psychographics	Interests, lifestyle, digital behavior, goals, motivations.
Pain Points	The frustrations or obstacles the persona faces related to the problem statement.
Technology Use	Devices, apps, comfort with technology.
Goals & Motivations	What outcomes they want; why they'd use your solution.
Quote	1–2 sentences capturing their mindset or struggle.
Success Metric	How they'd measure improvement or success (time saved, money earned, stress reduced).

Example Personas (for an AgriTech project)

Primary Persona – “Amina the Farmer”

- **Age / Location:** 34, lives in Kisumu County
- **Occupation:** Smallholder maize farmer, 2 acres
- **Income:** KES 20,000–40,000/month (seasonal)
- **Pain Points:** Crop diseases wipe out 30–50% of yield yearly; lacks timely advice.
- **Technology Use:** Uses Android smartphone, WhatsApp, and Facebook. Internet mostly via mobile data.
- **Goals:** Increase yield, reduce losses, secure steady income.
- **Quote:** “By the time I notice my crops are sick, it’s already too late.”
- **Success Metric:** Reduction in yield loss and improved harvest within one season.

Secondary Persona – “Samuel the Agro-Officer”

- **Age / Location:** 45, works with county agriculture office
- **Occupation:** Field officer supporting 200+ farmers
- **Pain Points:** Cannot visit all farmers frequently; lacks real-time reports.
- **Technology Use:** Tablet + laptop for reporting; WhatsApp for farmer communication.
- **Goals:** Monitor farm conditions remotely, generate quick reports.
- **Quote:** “If farmers could report issues directly, I could prioritize field visits better.”
- **Success Metric:** Fewer farm visits needed while maintaining yield support quality.

Edge Persona – “Lucy the Urban Gardener”

- **Age / Location:** 27, Nairobi
- **Occupation:** Graphic designer, grows vegetables on apartment balcony
- **Pain Points:** Limited knowledge of crop cycles; small space.
- **Technology Use:** Constantly online, comfortable using mobile apps.
- **Goals:** Learn how to manage her plants efficiently.
- **Quote:** “I want to grow healthy veggies, but I have no idea when to water or fertilize.”
- **Success Metric:** Healthy plants and reduced trial-and-error time.

Persona Validation Checklist

- Built from real or observed data (not pure assumptions).
- Includes motivations and decision-making factors, not just demographics.
- Expresses *emotional context* — how they feel about the problem.
- Identifies *success criteria* the product can impact.
- Differentiates between user roles (end user, admin, influencer, buyer).

2) EARLY ADOPTER PROFILE

What is an Early Adopter?

Early adopters are the first 5–10 % of users who try new tech solutions *before the general public*.

They take risks, give feedback, and are key to generating traction and credibility.

How to Identify Early Adopters

Ask students to describe:

1. **Where to find them** — online communities, events, institutions, demographics.
2. **Why they’d try the product early** — pain severity, openness to new tech, or incentives.
3. **How to reach them** — direct contact, referral, partnership, pilot programs.

Early Adopter Profile Template

Field	Description
Segment Name	e.g., “University Students in Tech Clubs”
Pain/Need	What makes them feel the problem strongly.
Adoption Trigger	The motivation to try something new (saves time, looks cool, helps job, etc.).
Discovery Channel	Where they can be reached (social media, events, institutions).
Barriers to Adoption	Reasons they might hesitate (cost, complexity, trust).
Incentive to Adopt	Beta access, free credits, recognition, certification.

Example (AgriSmart app)

- **Segment:** Progressive farmers under county cooperative groups
- **Pain/Need:** Frequent crop diseases but lack expert guidance; eager to modernize.
- **Adoption Trigger:** County/NGO partnership introducing digital tools.
- **Discovery Channel:** Local farming WhatsApp groups, cooperative meetings, agriculture expos.
- **Barrier:** Limited smartphone literacy or fear of app complexity.
- **Incentive:** Free first-season trial + training support.

3) ACQUISITION HYPOTHESIS (HOW YOU’LL GET THEM)

An **acquisition hypothesis** explains how the team will *find, attract, and convert* their first users.

Template

“We believe that [early adopter segment] will try [product name] through [acquisition channel] because [reason they trust or find it compelling]. Success will be measured by [metric].”

Example:

“We believe that county cooperative farmers will adopt *AgriSmart* through local training sessions and WhatsApp referrals because they already trust cooperative officers. Success will be measured by 200 app installs and 100 active users within 3 months.”

Example Acquisition Channels

Channel	Description	When to Use
Referral/Word of Mouth	Leverage early users to invite others; incentivize sharing.	Low budget MVP launches
Institutional Partnerships	Collaborate with NGOs, universities, county offices.	B2B or community impact projects
Content Marketing	Social posts, short explainer videos, blogs.	Educating new markets
Campus Outreach / Hackathons	For student-focused tech tools.	Youth innovation or edu-tech
Paid Ads	Social media campaigns.	Scalable consumer apps
Influencer/Advocate Marketing	Industry experts endorse solution.	Building credibility early

3) Define the Target Audience / Market

Objective: Show exactly who will use the product and why — not vague market buzzwords.

Deliverables

- Defined **primary** and **secondary** audience segments.

- 2–3 **user personas** (one primary, one secondary, one edge case).
- TAM / SAM / SOM estimates with assumptions (total addressable market, serviceable available market, serviceable obtainable market).
- Early adopter profile & acquisition hypothesis.

Persona template (fillable)

- **Name / Role:**
- **Age / Location / Socioeconomic:**
- **Behaviors & tech comfort:** (smartphone, internet, preferred channels)
- **Goals & motivations:**
- **Pain points:**
- **How they discover solutions:**
- **Quote:** (a one-line real or composite quote)

Example Persona (brief):

- Name: Amina, 34, smallholder farmer, uses feature phone + occasional smartphone, shops locally, motivated to increase yield, pain: no local agronomy advice.

Market sizing (how to do it)

- **TAM:** broad global or regional market size with source/assumption.
- **SAM:** portion your product addresses (e.g., smartphone users in East Africa).
- **SOM:** realistic initial share in 1–3 years with channel assumptions.

(If exact numeric data unavailable, state assumptions transparently: e.g., “Assume 40% smartphone penetration; targeting 10% of farmers in county X.”)

Early adopter & go-to-market hypothesis

- Where to find them (online communities, university labs, trade groups, clinics).
- Onboarding channel (partnerships, direct sales, app stores).
- Customer acquisition cost (CAC) estimate and payback logic (even high-level).

Validation checklist

- Personas are realistic and backed by research.
- Market assumptions are stated and justified.
- Early adopter channel is practically reachable within the project timeframe.

4) Define the Business & Revenue Model

Objective: Show how the product will be economically sustainable and attractive to investors.

Deliverables

- Primary revenue model chosen + 2 alternative models.
- Pricing strategy and basic unit economics (LTV, CAC, margin assumptions).
- 3-year revenue projection (simple table) with key assumptions.
- Funding ask if seeking investment: amount, use of funds (milestones), expected runway.

Revenue model options (explain when to pick each)

- **Subscription (SaaS):** recurring access; good for ongoing value (analytics, remote services).
- **Freemium → Premium:** free basic, paid advanced; works to grow user base quickly.
- **Transaction / Commission:** take a percentage on marketplace transactions.
- **Licensing / B2B contracts:** sell to institutions (schools, clinics) with per-seat or site license.
- **Ads / Sponsorships:** viable if large free user base and ad inventory.
- **One-time purchase + paid upgrades.**

Unit economics (simple formulas)

- **CAC (Customer Acquisition Cost):** total marketing & sales spend / new customers acquired.
- **LTV (Customer Lifetime Value):** average revenue per user per time × expected lifetime.
- **Payback period:** CAC / (average monthly revenue per user).
- **Gross margin:** (Revenue – direct cost) / Revenue.

Example projection table (students must provide)

Year	Users (end)	ARPU (annual)	Revenue	Key Assumption
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1	1,000	\$10	\$10,000	Pilot in region, 5% conversion
2	5,000	\$12	\$60,000	Expand channels
3	20,000	\$12	\$240,000	Scale partnerships

(Students should fill with local currency and justify assumptions.)

Investment & ROI narrative

- **Ask:** amount needed and what milestones it achieves (e.g., scale to 5k users, hire engineer, run 6-month pilot).
- **Investor ROI:** show exit scenarios (acquisition, revenue share) and breakeven timeline.

Validation checklist

- Revenue model fits user behavior and market.
- Unit economics are coherent (LTV > CAC).
- Funding ask has clear uses and milestones.

5) Define the Technology Stack

Objective: Provide a clear, justifiable stack showing how the product will be built, deployed, and scaled.

Deliverables

- Tech stack table (frontend, backend, database, storage, hosting, authentication, APIs, analytics, CI/CD).
- Architecture diagram (one page) showing user → app → services → data flow.
- Security & privacy section (data collected, storage, access rules, compliance notes).
- Maintenance & operations plan (how to deploy, monitor, backup, handle incidents).

Tech stack decision guide (pick and justify)

- **Frontend:** React / Vue / Angular / Flutter / native? Choose based on target devices and team skills.

- *Justify:* e.g., React for web SPAs with wide community; Flutter for cross-platform mobile.
- **Backend:** Node.js / Django / Ruby on Rails / Go / Java — pick based on concurrency needs and team expertise.
- **Database:** PostgreSQL (relational) vs MongoDB (document). Use PostgreSQL if structured data & transactions matter.
- **Storage:** S3-style object storage for files, images; CDN for static assets.
- **Authentication:** JWT / OAuth2 / Firebase Auth depending on scale.
- **Hosting / Cloud:** Heroku / Vercel / AWS / DigitalOcean / GCP — justify by cost/scale.
- **APIs & Integrations:** Payment gateway, SMS provider, email service.
- **Machine learning (if any):** model type, inference location (server vs edge), dataset storage, performance targets.
- **DevOps / CI-CD:** automated tests, pipelines, containerization (Docker), orchestration (Kubernetes optional).

Architecture considerations (scalability & reliability)

- Design for **stateless** servers so you can add instances horizontally.
- Use **caching** (Redis) for frequent reads.
- Store user files in object storage + use a CDN for delivery.
- Implement **health checks** and error monitoring (Sentry).
- Database backups and migration plan.
- Secrets management (don't commit keys).

Security & compliance

- Encrypt data at rest and in transit (TLS).
- Store only necessary personal data; follow data minimization.
- If handling payments, follow PCI requirements via third-party processors.
- Consider GDPR/local data protection rules: state how users can request deletion.

Deliverable: one-page architecture diagram (requirements)

- Show components and data flow.
- Mark external services and where data is stored.
- Annotate scaling and security measures.

Validation checklist

- Stack matches the MVP scope and team skills.
- Architecture shows how to scale from 0 → 10k users.
- Security basics are addressed.

6) Define the Impact & Scalability

Objective: Show the measurable difference the product will make and the path to growing adoption.

Deliverables

- Impact statement (1–3 paragraphs) linking product features to user outcomes.
- KPIs & success metrics (baseline + target within 6/12/24 months).
- Scalability roadmap (technical + business) with 3 milestones.
- Social/ethical considerations and how negative consequences will be mitigated.

Impact metrics (examples)

- **User outcomes:** % reduction in time/cost, % improvement in accuracy, saved money, people served.
- **Engagement metrics:** DAU / MAU, retention at 7/30/90 days.
- **Business metrics:** conversion rate, ARPU, churn.
- **Operational metrics:** API latency, uptime, error rate.

Scalability roadmap (example)

- **Phase 1 (Pilot, 0–6 months):** Validate with 100–1,000 users, fix critical UX bugs.
- **Phase 2 (Local scale, 6–18 months):** Expand to additional regions/states, improve automation, hire ops.
- **Phase 3 (Regional/global, 18–36 months):** Multi-language, partnerships, enterprise sales channel.

Scalability tactics (technical & go-to-market)

- **Technical:** containerize app, autoscaling, use managed DB, implement monitoring & feature flags.

- **Go-to-market:** partnerships (NGOs, schools, clinics), referral incentives, localized content, channel sales.

Social & ethical risk assessment

- Identify possible harms (bias in AI, misuse of data, exclusion).
- Describe mitigation (diverse datasets, opt-in data sharing, transparent model accuracy).

Validation checklist

- Clear numeric KPIs with baseline and target.
- Feasible roadmap with milestones tied to funding/user targets.
- Ethical risks identified with mitigation steps.